

Lesson 9 Set-Theoretic Proofs (involve sets)

Prove :

(i) $A \subseteq B$

(ii) $A = B$

$A \subseteq B$: Every element in A is in B

$$\forall x, \boxed{x \in A \rightarrow x \in B}$$

$A = B$: $A \subseteq B$ and $B \subseteq A$
(double inclusion)

Prop 1 A : set of multiples of 4
 B : set of even integers

$$A \subseteq B.$$

Proof : Let $x \in A$. Then x is a multiple of 4, and $x = 4n$ for some $n \in \mathbb{Z}$.
So, $x = 4n = 2(2n)$. Since $2n$ is an integer, x is even. Hence,
 $x \in B$.

Prop 2 Let $S = \{2m - 3n : m, n \in \mathbb{Z}\}$,
then $S = \mathbb{Z}$.

Proof First we prove that $\boxed{S \subseteq \mathbb{Z}}$.

Let $x \in S$. Then $x = 2m - 3n$ for some $n, m \in \mathbb{Z}$, and x is an integer.

Hence, $x \in \mathbb{Z}$

Next, we prove that $\mathbb{Z} \subseteq S$

Let $x \in \mathbb{Z}$, then $x = 2(-x) - 3(x)$,

Since $-x$ is an integer, $x \in S$.

Basic Set Equivalences

- $A \cap B = B \cap A$ commutative

2nd way to prove $A=B$:

$$x \in A \iff x \in B$$

both directions simultaneously

$$x \in A \cap B \iff x \in A \text{ and } x \in B$$

$$\iff x \in B \text{ and } x \in A$$

$$(p \wedge q = q \wedge p)$$

$$\iff x \in B \cap A$$

- $(A \cap B) \cap C = A \cap (B \cap C)$ associative

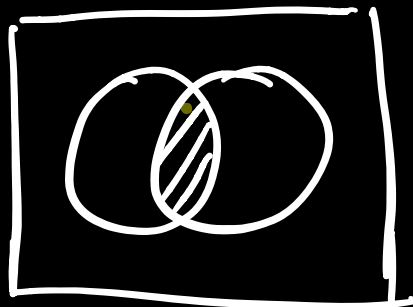
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ distributive

- $A \cap A = A, A \cup A = A$

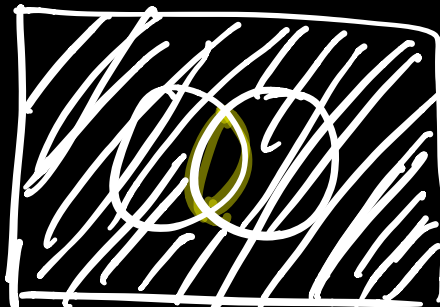
- De Morgan $\neg(p \wedge q) = \neg p \vee \neg q$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Ex3



$A \cap B$



$\overline{A \cap B}$



$\bar{A} \cup \bar{B}$:

everything in white or yellow

• "Intersection implies union"
("and" implies "or")

$$A \cap B \subseteq A$$

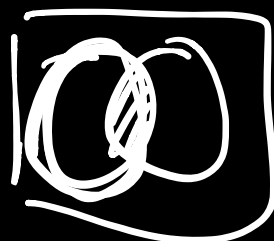
$$A \cap B \subseteq B$$

$$A \cap B \subseteq A \cup B$$

$$A \subseteq A \cup B$$

$$(P \rightarrow P) \text{ is true}$$

$$B \subseteq A \cup B$$



Prop 4 If $A \subseteq B$, then $\bar{B} \subseteq \bar{A}$.

Proof Let $x \in \bar{B}$. Then $x \notin B$, and $x \notin A$ (otherwise since $A \subseteq B$, $x \in A$ would imply $x \in B$).

Hence $x \in \bar{A}$. So, $\bar{B} \subseteq \bar{A}$.

Notes:
2nd proof: $x \in A \rightarrow x \in B$
= $x \notin B \rightarrow x \notin A$ contrapositive

Prop 5 $A \cap B = A \iff A \subseteq B$
iff

Proof ($\Rightarrow$) Since $A = A \cap B$, and $A \cap B \subseteq B$ (basic set equiv), we have $A \subseteq B$.

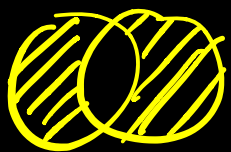
($\Leftarrow$) We need to prove that $A \cap B \subseteq A$ and $A \subseteq A \cap B$.

Since $A \cap B \subseteq A$ (always), we need to only prove $A \subseteq A \cap B$.

Let $x \in A$. Since $A \subseteq B$, $x \in B$.

So $x \in A$ and $x \in B$, and $x \in A \cap B$.

Prop 6 $A \Delta B = \underline{(A \cup B)} - \underline{(A \cap B)}$
 $(A - B) \cup (B - A)$



$$x \in A \Delta B \iff$$

$$\iff x \in (A \cup B) - (A \cap B)$$

Prop 7 If $A \subseteq B$, then $A \times C \subseteq B \times C$
Cartesian product

Proof Let $(x, y) \in A \times C$. Then
 $x \in A$ and $y \in C$. Since $A \subseteq B$,
 $x \in B$. So, $(x, y) \in B \times C$.